

Solutions to linear systems using homogeneous equations

Theorem

Suppose that $Ax = b$ is some consistent system of linear equations and x_0 is some solution. Then every solution to $Ax = b$ has the form $x_0 + y$ where y is a solution to the homogeneous equation $Ax = 0$.

$$A = \left(\begin{array}{cccc} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

↑
 x_2

↑
 x_4

free var.

$$x_1 = -2s + t$$

$$x_2 = s$$

$$x_3 = -2t$$

$$x_4 = t$$

Solve $Ax = 0$

for any numbers s, t .

General solution to $Ax = 0$.

Solve $Ax = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Form the aug.
matrix

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 2 \\ 0 & 0 & 1 & 2 & 3 \end{array} \right)$$

$$\begin{array}{l} x_1 = 2 \\ x_2 = 0 \\ x_3 = 3 \\ x_4 = 0 \end{array}$$

← set the
free variable
to 0.

General solution

$$x_1 = 2 - 2s + t$$

$$x_2 = s$$

$$x_3 = 3 - 2t$$

$$x_4 = t$$

$$\text{or } x = \begin{pmatrix} 2 \\ 0 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} -2s+t \\ s \\ -2t \\ t \end{pmatrix}$$

particular sol.

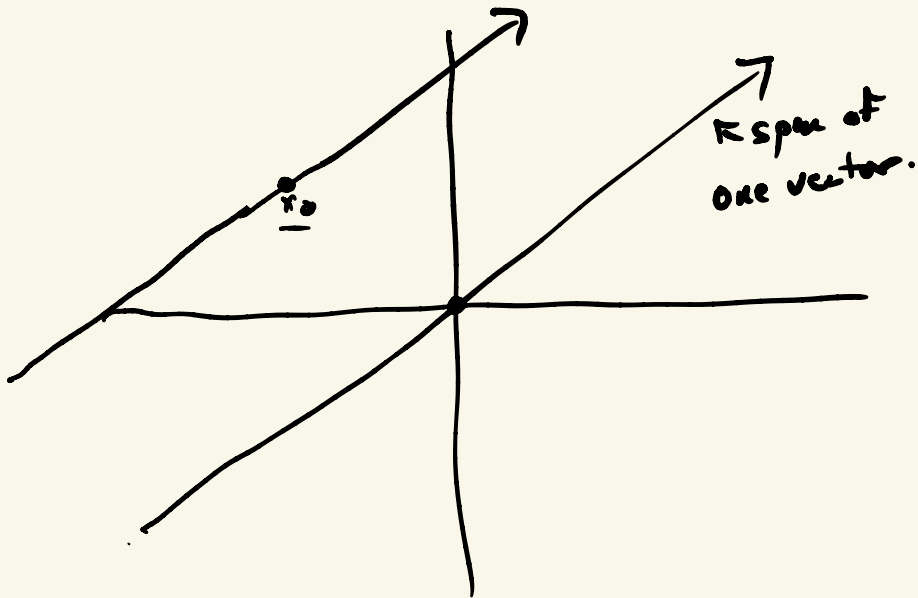
of $Ax = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Gen.
sol. of
 $Ax = 0$

$$\begin{pmatrix} -2s+t \\ s \\ -2t \\ t \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

this is all vectors spanned

by $\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$.



Linear independence

Definition

Suppose that $v_1, \dots, v_m$ are vectors in R^n and $A = (v_1 \dots v_m)$.

- We say that $v_1, \dots, v_m$ are **linearly independent** if whenever

$$\lambda_1 v_1 + \dots + \lambda_m v_m = 0$$

then $\lambda_1 = \dots = \lambda_m = 0$.

- That is, $v_1, \dots, v_m$ are linearly independent if the only solution to $Ax = 0$ is $x = 0$.
- We say that $v_1, \dots, v_m$ are **linearly dependent** if $Ax = 0$ has a non-trivial solution.

That is, if the only way to take a linear combination of $v_1, \dots, v_m$ and get 0 is to use only 0 scalars then $v_1, \dots, v_m$ are linearly independent.

Ex:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$v_1 \quad v_2 \quad v_3$

Can we solve

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

non-trivially
is, with
 $x \neq 0$.

The only solution here is $x = 0$

So v_1, v_2 and v_3 are lin. indep.

Ex. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \leftarrow$ are these lin indep?

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{array}{l} x_1 = 0 \\ x_2 = 0. \end{array}$$

In general, if $v_1, \dots, v_m$ are lin indep then any subset is also lin. indep.

Ex:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

v_1

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

v_2

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

v_3

$$\underline{v_3 = v_1 + v_2.}$$

or

$$\underline{v_1 + v_2 - v_3 = 0}$$

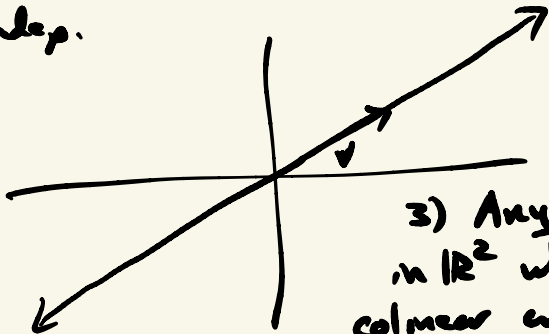
← demonstrates
that v_1, v_2 and v_3
are lin. dep.

Ex: In $\mathbb{R}^2$

1) $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is lin. dep. all by itself!

For instance $\frac{\pi}{2} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

2) If $v \in \mathbb{R}^2$, $v \neq 0$ then v is lin. indep.

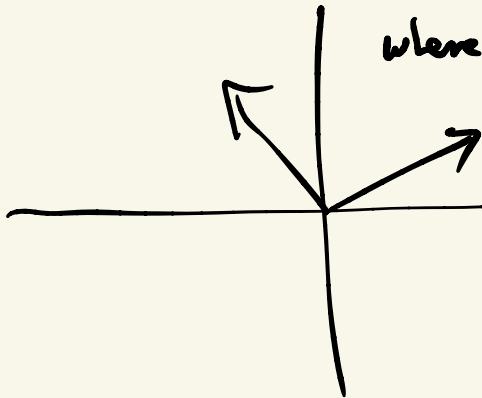


3) Any two vectors in $\mathbb{R}^2$ which are not colinear are lin. indep.

Can we have 3 vectors in $\mathbb{R}^2$ which are lin. indep?

Form $Ax = 0$

where $A = (v_1, v_2, v_3)$.



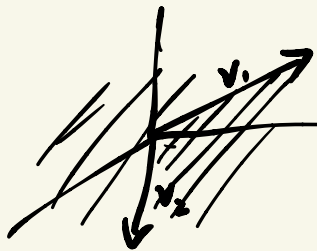
↑
Always have

some $x \neq 0$
as a solution.

Ex: In $\mathbb{R}^3$, 0 is lin. dep all by itself.

Any $v \neq 0$ will be lin. indep

What about 2 vectors? Any 2 non-colin.
vectors in $\mathbb{R}^3$ are lin indep.



Anything on the
plane spanned
by v_1 and v_2
is dep. on v_1 and v_2 .

If v_1 and v_2 are not colin. and
 v_3 is not on the plane gen. by v_1 and v_2
then v_1, v_2 and v_3 are lin. indep.

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We can't get 4 lin indep vectors in $\mathbb{R}^3$.

Ex: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftarrow$ lin. dep.
 $v_1 \quad v_2 \quad v_3$

v_1, v_2 are lin indep. v_2, v_3 are lin. indep.
 v_1, v_3

The size of linearly independent sets

Theorem

If $v_1, \dots, v_m$ are in R^n and $m > n$ then $v_1, \dots, v_m$ are linearly dependent.

If $v_1, \dots, v_m$ are linearly dependent then at least one of these vectors can be written as a linear combination of the others.